ABHYUDIT ADHIKARI

AI/ML Engineer | LLM Systems & Computer Vision

+977 9748284848 • abhyudit.adhikari@gmail.com • GitHub • Portfolio • Kathmandu, Nepal (Open to remote and relocation)

SUMMARY

AI/ML Engineer who ships LLM systems to production: cut inference costs ~90% on a live shipping-industry AI platform, built vision-based document extraction, and deployed a 95%-accuracy classifier for an award-recognized AI startup. IEEE-published.

EDUCATION

Advanced Engineering Program in **AI Agent Workflows & Agentic Systems — IIT Madras Pravartak** Apr 2026 – Nov 2026 (expected)

Coursework: *multi-agent collaboration topologies, RAG and context engineering (pgvector, hybrid retrieval), multi-agent orchestration and observability (LangChain, LangGraph, LangSmith).*

Bachelor of Engineering: **Computer Engineering (CGPA: 3.60/4.0) — Kathmandu University** Feb 2022 – Jul 2026

EXPERIENCE

Co-founder & AI Lead — CreoTech (Part-time) May 2026 – Present

- Led AI and engineering for the MVP of a B2B employee-wellbeing platform: a webcam-based posture estimation tool (MediaPipe landmarks and a regression model) with eye-strain reminders and guided exercises (Python, FastAPI, Next.js, OpenCV).

Software Engineer (Traineeship) — Infinite Computer Solutions (Full-time) Aug 2026–Oct 2026

- Built an AI CV-screening platform that matches a pool of 1,000+ CVs against multiple job descriptions at once, surfacing cross-matches (candidates who fit several roles) and shortlisting top candidates per JD. More details given below under projects.

Software Engineer — NAVIO (Full-time) Oct 2025 – Feb 2026

- Cut LLM costs ~90% by migrating a shipment email-parsing system from OpenAI to Gemini, using dynamic model routing, input sanitization, Redis caching of suggested questions, and atomic DB locking to prevent duplicate inference and race conditions.
- Built B/L extraction for Pontus, a live production shipping chatbot: Gemini vision with per-field bounding boxes on complex maritime B/Ls.
- Implemented role-adaptive prompts for freight forwarders, cargo owners, and charterers, and a fault-tolerant regeneration workflow (soft deletes, rollback) that prevents data corruption.

Co-founder & ML Engineer — Dr. Fish (Part-time) Feb 2024 – Nov 2025

- Trained a TensorFlow ResNet50 fish-disease classifier to 95% accuracy and deployed it on Hugging Face for on-demand inference, reducing latency from 200 ms to 120 ms.
- Built the FastAPI backend that served the model to the user-facing app. Finalist, ICT Award 2024; Top 7, DIAL accelerator program (Swiss Agency for Development and Cooperation).

PUBLICATION

Real-Time Assistive Navigation for Visually Impaired Using YOLOv12 and MediaPipe Pose Estimation GitHub • Paper

Co-authored research on a real-time assistive navigation system for the visually impaired. Built the system integrating YOLOv12 object detection and MediaPipe pose estimation that allows users to locate indoor objects via voice commands and spatial audio feedback, achieving 94.7% detection accuracy with 1.2s response latency.

IEEE Xplore, 8th International Conference on DSIT 2025 - ***DOI:*** *10.1109/DSIT67006.2025.11389902*

PROJECTS

AI CV Screening System GitHub

- Built a four-stage CV screening platform with PDF OCR, eligibility rules, and hybrid pgvector/full-text retrieval; fused and reranked evidence to shortlist up to 15 candidates per job. (*FastAPI, Angular, PostgreSQL/pgvector, Redis, Celery, Docker, SentenceTransformers, Gemini, pytest)*
- Added citation-verified LLM assessments, versioned scoring, human-review routing; built recoverable Celery workflows with quota controls.
- Cut estimated LLM input tokens by 61.8% across 12 offline cases through evidence-preserving prompt compaction.
- Built GitHub Actions CI covering tests, migrations, and Docker image builds.

Brain Tissue Classification using Hyperspectral Imaging (ongoing research)

- Implemented **HybridSN** with spectral attention for brain tissue classification, achieving **94% F1-score**. Designed a **data pipeline** for efficient HSI calibration, normalization, and patch extraction, handling complex datasets.

SKILLS

Languages: Python, SQL, TypeScript

AI/ML: PyTorch, TensorFlow, OpenCV, MediaPipe, YOLO, Hugging Face, RAG, pgvector, Gemini/OpenAI APIs, scikit-learn, NumPy, pandas

Backend & Infra: FastAPI, PostgreSQL, Redis, Docker, Next.js, Redis, GitHub Actions CI, pytest